Java Web Application Deployment on Azure Virtual Machine

Project Overview

This project demonstrates the end-to-end deployment of a Java web application on a cloud environment using **Microsoft Azure**. It involves creating cloud infrastructure, configuring networking, installing required software, building the application using Maven, and hosting it on a virtual machine. The deployed application is made accessible over the internet using the VM's public IP address.

Aim

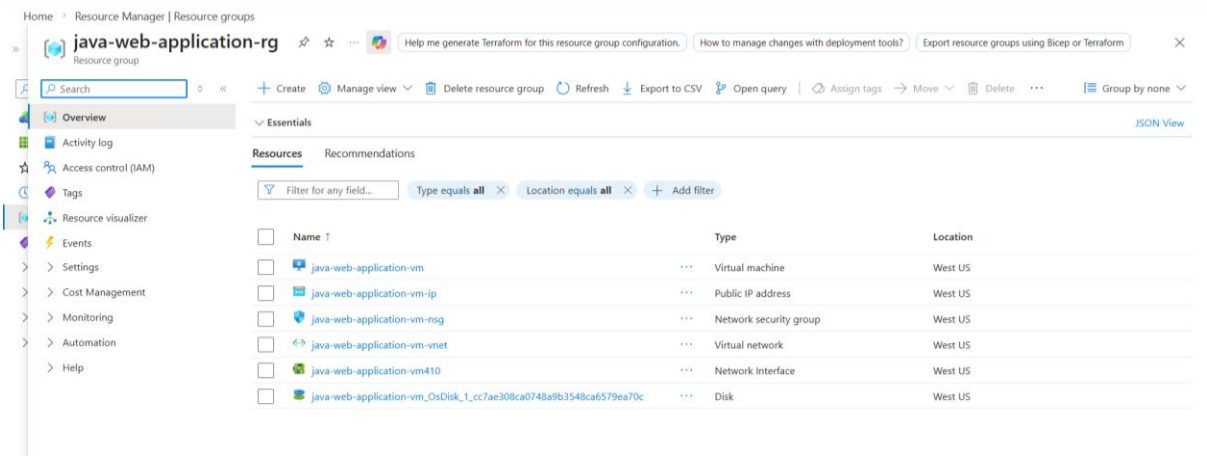
To deploy and run a Java-based web application on an Azure Virtual Machine by setting up the required environment, building the project, and enabling public access.

Tools & Technologies Used

- **Java (JDK 11)** – Application runtime
- **Apache Maven** – Build and dependency management
- **Linux (Ubuntu VM)** – Hosting environment
- **Git & GitHub** – Source code repository
- **Azure Virtual Machine** – Application hosting
- **Azure Resource Manager** – Resource management
- **Network Security Groups (NSG)** – Traffic control
- **SSH (Port 22)** – Secure remote access

Step 1: Create Resource Group

Created a resource group named **java-web-application-rg** to organize and manage all Azure resources related to this project.



Step 2: Create Virtual Machine

Provisioned a Linux-based **Azure Virtual Machine** named **java-web-application-vm** to host the application.

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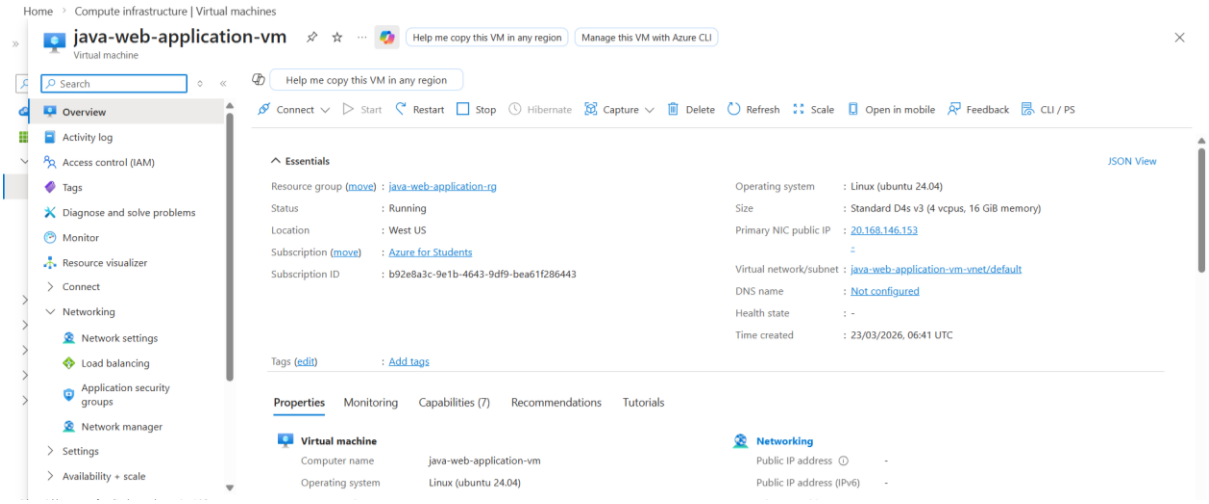
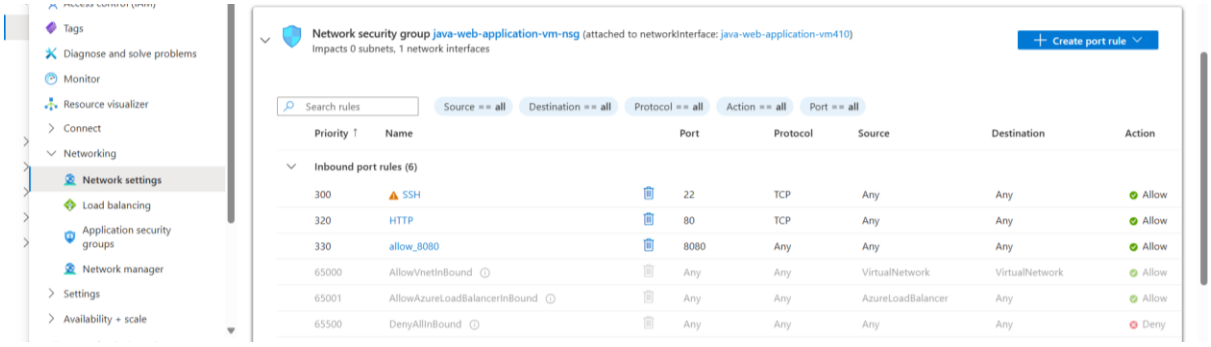


Step 3: Configure Inbound Security Rules

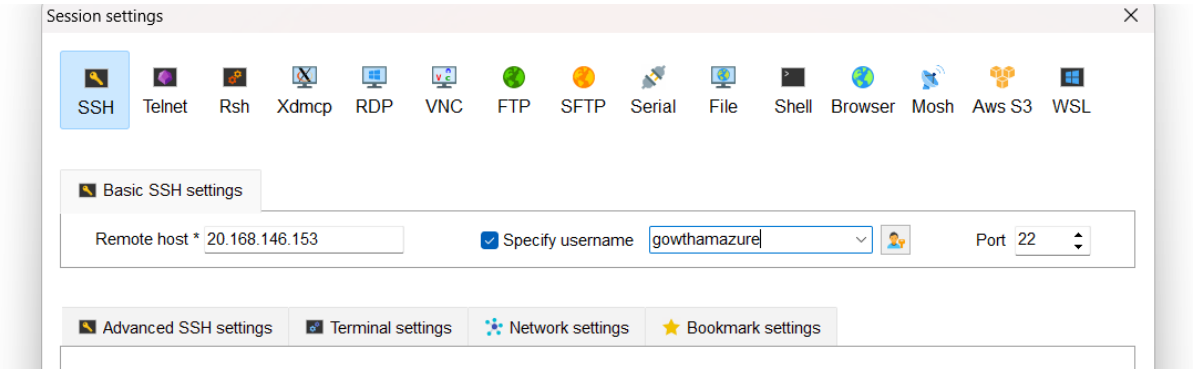
Configured networking by allowing the following inbound ports:

- **22 (SSH)** – Remote access
- **80 (HTTP)** – Web traffic
- **8080** – Application access



Step 4: Connect to VM via SSH

Connected securely to the virtual machine using SSH:



Step 5: Clone Git Repository

Cloned the Java application source code from GitHub:

git clone <https://github.com/gowthamsaikadali/boardgame.git>

```
root@java-web-application-vm:/home/gowthamazure# git clone https://github.com/gowthamsaikadali/boardgame.git
Cloning into 'boardgame'...
remote: Enumerating objects: 70, done.
remote: Counting objects: 100% (70/70), done.
remote: Compressing objects: 100% (48/48), done.
remote: Total 70 (delta 11), reused 70 (delta 11), pack-reused 0 (from 0)
Receiving objects: 100% (70/70), 53.25 KiB | 2.54 MiB/s, done.
Resolving deltas: 100% (11/11), done.
root@java-web-application-vm:/home/gowthamazure#
```

cd boardgame

```
Resolving deltas: 100% (11/11), done.
root@java-web-application-vm:/home/gowthamazure# ls
boardgame
root@java-web-application-vm:/home/gowthamazure# cd boardgame/
root@java-web-application-vm:/home/gowthamazure/boardgame# ls
Dockerfile  Jenkinsfile  Jenkinsfilekrishna  README.md  deployment-service.yaml  mvnw  mvnw.cmd  pom.xml  sonar-project.properties  src
root@java-web-application-vm:/home/gowthamazure/boardgame#
```

Step 6: Install Java (JDK 11)

Installed Java to provide the runtime environment required to run the application:

apt update

```
root@java-web-application-vm:/home/gowthamazure/boardgame# apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1841 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [178 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1569 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [320 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [33.1 kB]
Get:21 http://azure.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2825 kB]
```

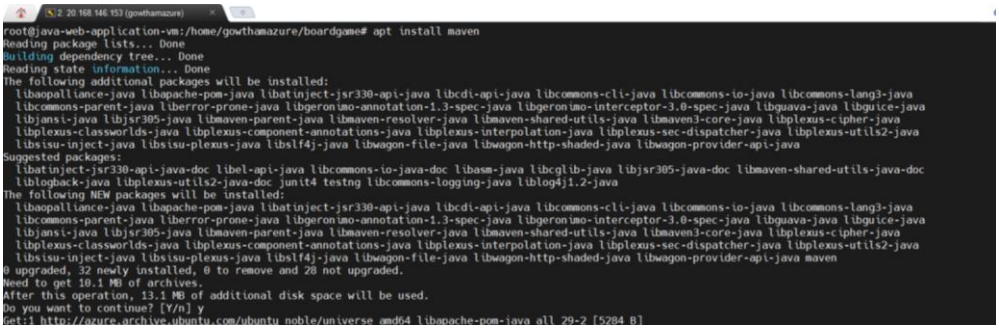
apt install openjdk-11-jre-headless -y

```
root@java-web-application-vm:/home/gowthamazure/boardgame# apt install openjdk-11-jre-headless
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java fontconfig-config fonts-dejavu-core fonts-dejavu-mono java-common libasound2-data libasound2t64
  libavahi-client3 libavahi-common-data libavahi-common3 libcups2t64 libfontconfig1 libgraphite2-3 libharfbuzz0b libjpeg-turbo8 libjpeg8 liblcms2-2
  libpcsclite1
Suggested packages:
  default-jre alsa-utils libasound2-plugins cups-common liblcms2-utils pcscd libnss-mdns fonts-dejavu-extra fonts-ipafont-gothic fonts-ipafont-mincho
  fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic
The following NEW packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java fontconfig-config fonts-dejavu-core fonts-dejavu-mono java-common libasound2-data libasound2t64
  libavahi-client3 libavahi-common-data libavahi-common3 libcups2t64 libfontconfig1 libgraphite2-3 libharfbuzz0b libjpeg-turbo8 libjpeg8 liblcms2-2
  libpcsclite1 openjdk-11-jre-headless
0 upgraded, 21 newly installed, 0 to remove and 28 not upgraded.
Need to get 45.6 MB of archives.
After this operation, 187 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 alsa-topology-conf all 1.2.5.1-2 [15.5 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libasound2-data all 1.2.11-1ubuntu0.2 [21.3 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libasound2t64 amd64 1.2.11-1ubuntu0.2 [398 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 alsa-ucm-conf all 1.2.10-1ubuntu5.9 [67.8 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 ca-certificates-java all 20240118 [11.6 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 fonts-dejavu-mono all 2.37-8 [502 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 fonts-dejavu-core all 2.37-8 [835 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 fontconfig-config amd64 2.15.0-1.1ubuntu2 [37.3 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 java-common all 0.75+exp1 [6798 B]
Get:10 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libavahi-common-data amd64 0.8-13ubuntu6.1 [29.9 kB]
Get:11 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libavahi-common3 amd64 0.8-13ubuntu6.1 [23.4 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libavahi-client3 amd64 0.8-13ubuntu6.1 [26.8 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 libcups2t64 amd64 2.4.7-1.2ubuntu7.9 [273 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu noble/main amd64 libfontconfig1 amd64 2.15.0-1.1ubuntu2 [139 kB]
```

Step 7: Install Maven

Installed Apache Maven to build and package the application:

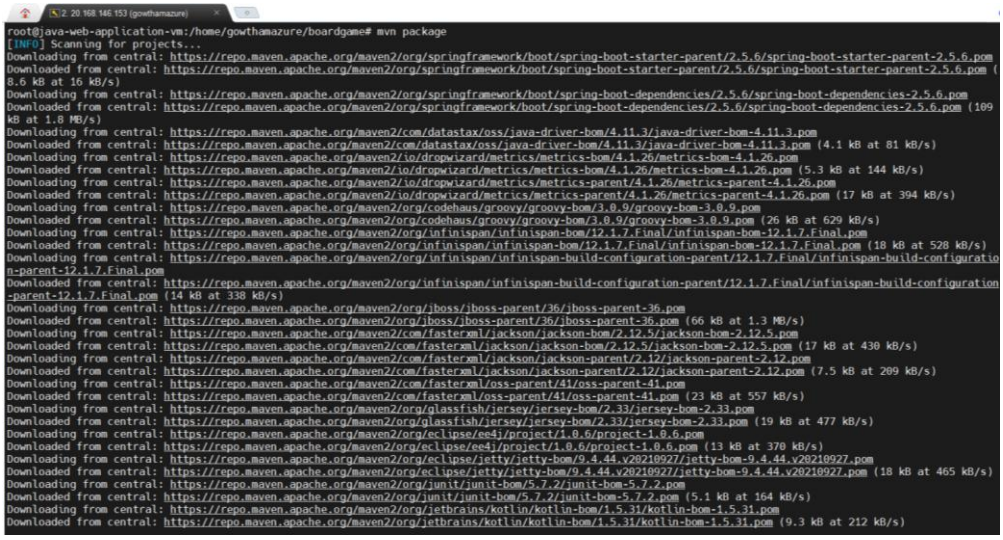
apt install maven -y



Step 8: Build the Application

Built the application using Maven, which generates a JAR file in the target directory:

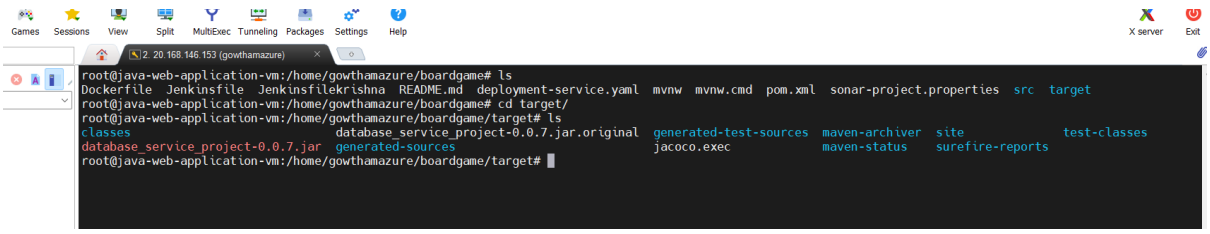
mvn package



9: Navigate to Target Folder

Accessed the target directory and verified the generated JAR file:

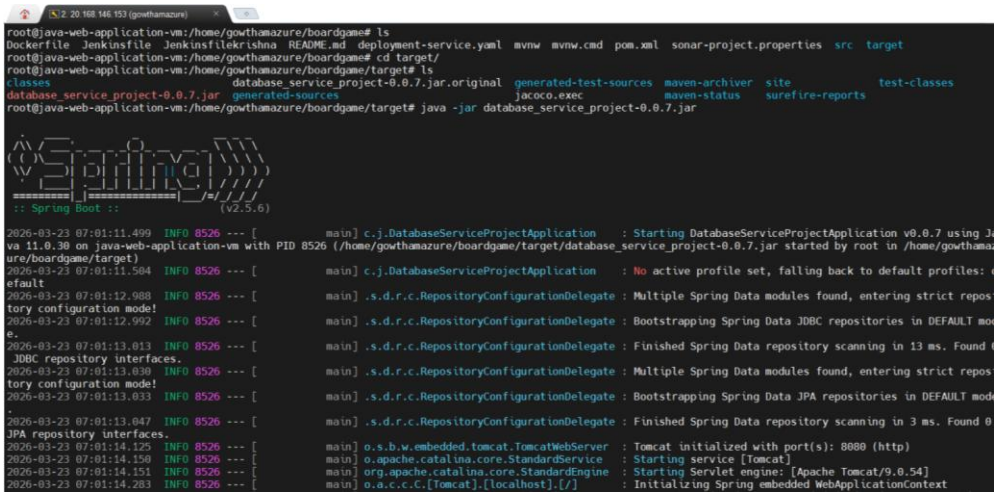
cd target
ls



Step 10: Run the Application

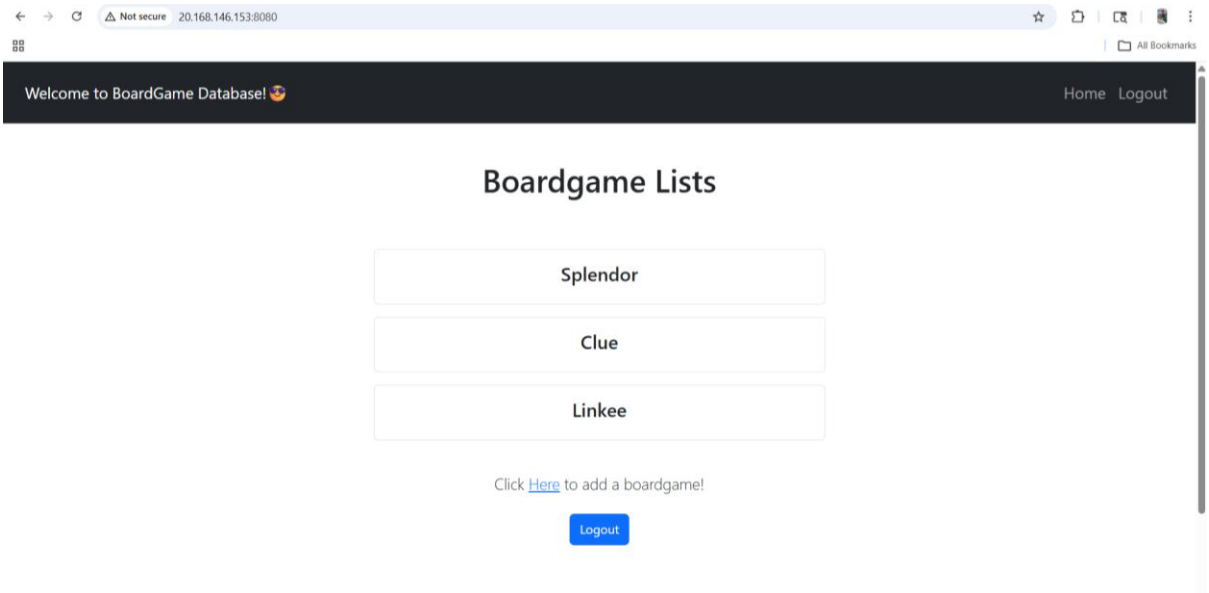
Executed the JAR file to start the application:

java -jar database_server_project-0.0.7.jar



Step 11: Access Application in Browser

Opened the application in a web browser using:
http://<VM-Public-IP>:8080



Output

- Java application successfully built using Maven
- Application deployed and running on Azure VM
- Network configured to allow external access
- Application accessible via public IP on port 8080

Conclusion

This project demonstrates the successful deployment of a Java web application on **Microsoft Azure**. It provides hands-on experience with virtual machines, networking, and application deployment. The project strengthens core skills in cloud computing and DevOps, including environment setup, build automation, and real-time application hosting.